

Review of Traps from Last Semester (2): Graphs and Equations

after half term trap full review · 70 minutes · 1-to-3 Online Lesson

Review scope: Hall 3-6 (area: parallelogram, triangle, trapezoid, polygon) + Hall 14-15 (simple equations)

Core traps: □ T4 Area Trap + T5 Graph Trap + T3 Equation Trap + T7 Comprehensive Trap

● SSPA compulsory exam

SSPA related: ● High frequency The score for Paper 1 is about 25-30%, graphs and equations are the key to score for Paper 2

Prerequisite knowledge: Hall 3-4 (area formula) · Hall 5 (area trap special subject) Lesson 6 (Comprehensive) · Lesson 14-15 (Equations)

Goals of this lesson: ❶ Comprehensive review of area traps (bottom-height confusion, forgetting $\div 2$, unit error) ❷ Comprehensive review of equation traps ❸ Comprehensive application of graphs + equations

Student Name:

Class:

Date:

Time Spent:

I.Warm-Up Questions (total 5 question , 5 minutes)

#	Question	Difficulty	Working Space (Show full working)
1	What is the formula for the area of a parallelogram?	Basic	
2	What is the formula for the area of a triangle?	Basic	
3	What is the formula for the area of a trapezoid?	Basic	
4	Solve the equation: $3x = 15$, $x = ?$	Basic	
5	Solve the equation: $x + 7 = 20$, $x = ?$	Basic	

II.Core Knowledge + trapreview

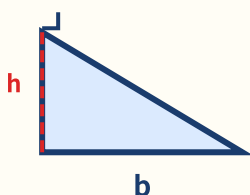
Knowledge point 1: General review of area trap ● T4+T5 trap

Comprehensive review of T4 area formula traps:

- ① **Triangle forgets $\div 2$ |||SEP|||: $\text{base} \times \text{height} \div 2$, not $\text{base} \times \text{height}$! (Confused with parallelogram) - SSPA's highest frequency loss point** Trapezoid forgets $\div 2$ |||SEP|||: $(\text{upper base} + \text{lower base}) \times \text{height} \div 2$, not $(\text{upper base} + \text{lower base}) \times \text{height}$!
- ② **The base and the height do not correspond |||SEP|||: The height must be perpendicular to the base, and you cannot just take a hypotenuse as the height** The height of the parallelogram is not the hypotenuse |||SEP|||: Height = vertical distance, hypotenuse $\neq$ height
- ③ **Unit error |||SEP|||: The area unit is cm^2/m^2 , not cm/m** Compound graphics disassembly error |||SEP|||: Overlapping or omission when disassembling into basic graphics
- ④ **The height of a parallelogram is not the hypotenuse:** Height = vertical distance, hypotenuse $\neq$ height
- ⑤ **unit error:** The unit of area is cm^2/m^2 , not cm/m
- ⑥ **Composite graphics disassembly error:** Overlap or omission when split into basic graphics

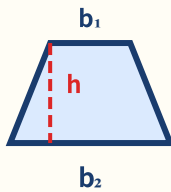
□ SVG trap diagram 1: Triangle - the corresponding relationship between base and height (T4 highest frequency trap)

The triangle below has base = 10 cm and height = 6 cm (vertical). Area = ?



□ **SVG Trap Illustration 2: Trapezoid - Forget $\div 2$ (T4 Second High Frequency Trap)**

The trapezoid in the figure below has upper base = 4 cm, lower base = 8 cm, and height = 5 cm. Area = ?



🧠 **Tip: "A triangular trapezoid must be divided by two, and the parallel four sides must not be divided; the height should be vertical, not oblique, and the unit should be remembered to be square."**

⚠️ **T4 Highest Frequency Trap: Triangles and Trapezoides Forget $\div 2$! Both graph area formulas have " $\div 2$ "!**

⚠️ **T5 graphics trap: When dismantling composite graphics, make sure there is no overlap or omission, and the area of each part after disassembly is added up.**

Knowledge Point 1 Synchronization practice

#	Question	Difficulty	Working Space
Example 1	Triangle base = 12 cm, height = 8 cm. Area = ?	🌱	
Example 2	Trapezoid upper base = 6 cm, lower base = 10 cm, height = 4 cm. Area = ?	🌿	

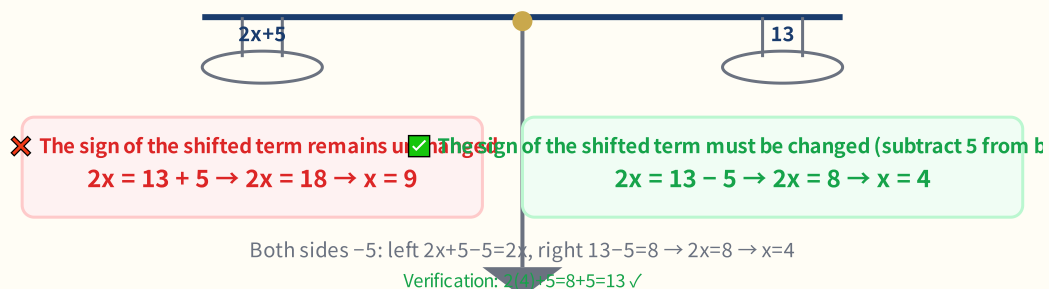
Knowledge point 2: General review of equation traps ● T3 trap

Comprehensive review of T3 equation traps:

- ① **Transfer terms and forget to change signs** : $x + 3 = 7 \rightarrow x = 7 + 3 = 10$ (wrong! It should be $x = 7 - 3 = 4$)
- ② **Both sides are out of sync |||SEP|||: Equations must be "equivalent axioms" - do the same operation on both sides** Error in removing brackets
- ③ **+ 3 (the negative sign must be multiplied by each term)** : $-(2x-3) = -2x$ Coefficient simplification error
- ④ **$= 8 \times 2 = 16$ (wrong! It should be : $2x = 8 \rightarrow x$** Forgot to check the root |||SEP|||: Substitute into the original equation to check, which can avoid 80% of calculation errors $x = 8 \div 2 = 4$)
- ⑤ **Forgot to check root**: Substituting into the original equation for checking can avoid 80% of calculation errors

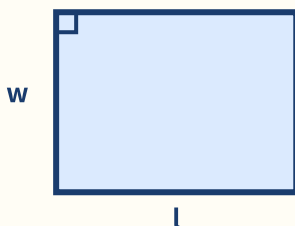
□ SVG Trap Diagram 3: Equation - Shifting Terms and Changing Signs (T3 highest frequency trap, SSPA must be tested every year)

Solve the equation: $2x + 5 = 13$, find x °



□ SVG Trap Illustration 4: Graph + Equation Synthesis—Assume unknowns to find height (T4+T3 mixed trap)

Area of parallelogram = 48 cm^2 , base = 8 cm , let height be $h \text{ cm}$. beg h °



🧠 **Tip:** "Add and subtract both sides of the equation at the same time, and the sign of the transfer term must be reversed; substitute the area formula into it, and use the equation to solve the unknown number."

⚠ T3 highest frequency: forgetting to change the sign when moving items - addition changes to subtraction, subtraction changes to addition! Do the same operation on both sides!

⚠ T4+T3 Mixed Trap: For graphing questions, write the area formula first, then do algebra, and finally solve the equation. Don't skip a step!

Knowledge Point 2 Synchronization practice

#	Question	Difficulty	Working Space
Example 3	Solve the equation: $4x - 3 = 17$, $x = ?$		
Example 4	Solve the equation: $2(x + 3) = 14$, $x = ?$		

Knowledge point 3: Comprehensive application of graphics + equations

T4+T5+T3+T7 mixed

Graph + Equation Comprehensive Problem Solving Framework (SSPA Paper 2 pull question):

- ① **Drawing marks** |||SEP|||: Mark the known and unknown lengths on the graphWrite the area formula |||SEP|||: Write the correct formula according to the type of graph (note $\div 2$!)
- ② **Assume unknowns**: Set the unknown quantity in the formula to
- ③ **Substitute the known number**: Substitute the known area, base, height, etc. into
- ④ **Solve the equation**: Use the equivalent axiom to find out
- ⑤ **Verify**: Substitute back to the original formula to check whether the area is consistent
- ⑥ **verify**: Substitute back to the original formula to check whether the area matches

①

Drawing markup

Label known and unknown lengths

②

write formula

Correct area formula ($\div 2$?)

③

Let the unknowns

The unknown quantity is set to x

④

Solving Equations Algebraically

Substitute $\rightarrow$ Solve $\rightarrow$ Verify

Example 5 - Find the height of a triangle (T4+T3 mixed)

Area of triangle = 36 cm^2 , base = 9 cm . Let the height be $h \text{ cm}$, find h .

✗ Common mistakes

$$9 \times h = 36 \rightarrow h = 4$$

Forgot that the formula for the area of a triangle is $\div 2$!
It should be $9 \times h \div 2 = 36$

✓ Correct solution

$$\frac{1}{2} \times 9 \times h = 36 \rightarrow 9h = 72 \rightarrow h = 8 \text{ cm}$$

Formula: base $\times$ height $\div 2 =$ area $\rightarrow 9h \div 2 = 36 \rightarrow 9h = 72 \rightarrow h = 8$

Example 6 - Finding the upper and base of a trapezoid (T4+T3 mixed)

Trapezoid area = 40 cm^2 , base = 12 cm, height = 5 cm. Assume that the upper bottom is $a \text{ cm}$, find a .

⚠ T7 Comprehensive Trap: Mixed questions of graphs and equations. The most common mistake is writing the area formula incorrectly (forgetting $\div 2$), resulting in all the equations being wrong!

III. Lesson Layered Synchronization Practice

Basic layer (total 7 questions, all must do — formula basic)

#	Question	Difficulty	Working Space
6	Base of parallelogram = 9 cm, height = 6 cm. Area = ?		
7	Triangle base = 14 cm, height = 5 cm. Area = ?		
8	Trapezoid upper base = 5 cm, lower base = 9 cm, height = 6 cm. Area = ?		
9	Solve the equation: $x + 8 = 15$, $x = ?$		
10	Solve the equation: $5y = 35$, $y = ?$		
11	Solve the equation: $x - 6 = 9$, $x = ?$		
12	Area of triangle = 20 cm^2 , base = 5 cm. High = ?		

Advanced layer (total 7 questions,   choose do — trap identification and mixed question)

#	Question	Difficulty	Working Space
13	Area of parallelogram = 54 cm^2 , base = 9 cm. High = ?		
14	Solve the equation: $3x + 4 = 19$, $x = ?$		
15	Solve the equation: $2x - 5 = x + 3$, $x = ?$		
16	Trapezoid area = 48 cm^2 , upper base + lower base = 12 cm. High = ?		
17	The base of a triangle is 8 cm and the height is 1.5 times the base. Area = ?		
18	Solve the equation: $2(x - 3) + 4 = 12$, $x = ?$		
19	Parallelograms and triangles have equal bases and equal heights. Area of parallelogram = 32 cm^2 . Area of triangle = ?		

 challenge layer (total 6 questions,  choose do — graphics + equationcomprehensive)

#	Question	Difficulty	Working Space
20	Area of triangle = 30 cm^2 , height = 5 cm. Let the base be b cm, and use the equation to find b .		
21	The upper base of a trapezoid is 4 cm shorter than the lower base, lower base = 10 cm, area = 56 cm^2 . Let the height be h cm, find h .		
22	Solve the equation: $\frac{x}{2} + 3 = \frac{x}{3} + 4$, $x = ?$		
23	The rectangle is 12 cm long. If the width increases by 2 cm, the area increases by 24 cm^2 . It turns out that $Kuo = ?$		
24	A composite figure consists of a square (side length 6 cm) plus a triangle (base = 6 cm), total area = 54 cm^2 . Height of triangle = ?		
25	The lower base of a trapezoid is 3 times the upper base, upper base = x cm, height = 4 cm, area = 64 cm^2 . Find the length of the top and bottom.		

IV. application questions special topic training (total 6 questions)

Graphic application question (all must do, column formula → calculate → answer sentence)

#	Question	Difficulty	Working Space
26	A triangular garden bed has a base length of 8 meters and a height of 5 meters. What is the area of the garden in square meters?		
27	The parallelogram grass base is 12 meters long and 7 meters high. How many square meters is the area?		
28	The terraced field is 6 meters above ground, 10 meters below and 4 meters high. What is the total area of the fields?		
29	A triangular billboard has an area of 24 square meters and a base of 8 meters. How many meters high is it? (column of equations)		

30	The rectangular garden is 15 meters long and wide $\frac{2}{3}$. What is the garden area?		
31	The upper bottom of the trapezoidal swimming pool is 8 meters, the lower bottom is 1.5 times that of the upper bottom, and the area is 80 square meters. How many meters high is the swimming pool? (column of equations)		

V. Ultimate challenge area (total 3 question)

#	Question	Difficulty	Working Space
 1	A polygon consists of a parallelogram and a triangle, both having equal bases (10 cm). The parallelogram is 6 cm high and the triangle is 4 cm high. Total area = ?		
 2	Rectangle perimeter = 36 cm. The length is 2 times the breadth. Assume the width is w cm, make equations to find the length and width, and then find the area.		
 3	Trapezoid area = 72 cm^2 , base = 14 cm, height = 6 cm. Let the upper base be a cm. (1) Use a series of equations to find a . (2) If this trapezoid is cut along the midline, the height of each small trapezoid is half of the original one. Find the area of each small trapezoid.		

VI. Class after homework

Basic must-do questions (total 5 questions)

#	Question	Difficulty	Working Space
H1	Triangle base = 10 cm, height = 6 cm. Area = ?		
H2	Trapezoid upper base = 7 cm, lower base = 13 cm, height = 5 cm. Area = ?		
H3	Solve the equation: $6x = 42$, $x = ?$		
H4	Solve the equation: $3x + 7 = 22$, $x = ?$		

H5	Area of parallelogram = 45 cm^2 , base = 9 cm. High = ?		
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Advanced choose doquestion (total 3 questions,  choose do)

#	Question	Difficulty	Working Space
H6	Area of triangle = 42 cm^2 , base = 12 cm. Let the height be h cm, and use the equation to find h .		
H7	Solve the equation: $2(x+5) - 3 = 15$, $x = ?$		
H8	Trapezoid area = 60 cm^2 , upper base + lower base = 15 cm. High = ?		

VII. The Lesson core common errors summary

 **Review of Learning Objectives - After completing this lesson you should be able to:**

☐ Identify all trap types in our hall

☐ Solve  basic questions independently (100% correct)

☐ Challenge  Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	Forget the area of a triangle $\div 2$ (T4)	Area of triangle = base $\times$ height $\div 2$
2	The area of the trapezoid is forgotten $\div 2$ (T4)	Trapezoidal area = (upper base + lower base) $\times$ height $\div 2$
3	Use hypotenuse as height (T5): Height of parallelogram $\neq$ hypotenuse	The height must be perpendicular to the base, not the length of the hypotenuse
4	Forgot to change the number when moving the item (T3) : $x+3=7 \rightarrow x=7+3$	The sign of the transfer needs to change: $x+3=7 \rightarrow x=7-3=4$
5	Bracket removal error (T3) : $-(2x-3)=-2x-3$	Multiply each term with a negative sign: $-(2x-3)=-2x+3$
6	Graphs + Equations: Wrong formula substitution (T7)	First write the correct formula $\rightarrow$ Substitute the known numbers $\rightarrow$ Assume the unknown $\rightarrow$ Solve the equation
7	Wrong unit of area SEP : write cm instead of cm^2 (T5): write cm instead of cm^2 (T5)	The area unit must be square: $\text{cm}^2, \text{m}^2, \text{km}^2$

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 Related topics: L14 Understanding algebraic expressions • L15 Equation word problems • L31 Advanced algebraic expressions